## 5-2b Computing the Price Elasticity of Supply

Now that we have a general understanding of the price elasticity of supply, let's be more precise. Economists compute the price elasticity of supply as the percentage change in the quantity supplied divided by the percentage change in the price. That is,

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}.$$

For example, suppose that an increase in the price of milk from $\$2.85$ to $\$3.15$ a gallon raises the amount that dairy farmers produce from $9{,}000$ to $11{,}000$ gallons per month. Using the midpoint method, we calculate the percentage change in price as

$$\text{Percentage change in price} = (3.15 - 2.85)/3.00 \times 100 = 10 \text{ percent}.$$

Similarly, we calculate the percentage change in quantity supplied as

$$\text{Percentage change in quantity supplied} = (11{,}000 - 9{,}000)/10{,}000 \times 100 = 20 \text{ percent}.$$

In this case, the price elasticity of supply is

$$\text{Price elasticity of supply} = \frac{20 \text{ percent}}{10 \text{ percent}} = 2.$$

In this example, the elasticity of $2$ indicates that the quantity supplied changes proportionately twice as much as the price.

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